EPOC X Hybrid Cursor Control System

This project is a complete, production-ready Brain-Computer Interface (BCI) pipeline designed for the **Emotiv EPOC X**. It enables hands-free 2D mouse cursor control using raw EEG (brainwave) intent, combined with a highly reliable EMG (muscle) "jaw clench" mechanism for clicking.

Unlike simple dwell-timer systems, this hybrid approach separates navigation (EEG) from activation (EMG), eliminating the "Midas Touch" problem and drastically improving usability.

🧠 Core Innovations

1. **Hybrid EEG + EMG Pipeline:** Uses spatial filtering (CSP) on the Mu/Beta bands for 2D cursor movement, while continuously monitoring temporal/frontal channels for high-frequency variance (jaw clenching) to trigger an OS-level mouse click.
2. **Adaptive Baseline Normalization:** Uses an Exponential Moving Average (EMA) to track your resting brain state. This automatically corrects for sensor drying, sweat, and natural neurological fatigue over long sessions.
3. **Kinematic Kalman Filter:** Injects simulated physics (mass and friction) into the cursor. Instead of jittering erratically, the cursor glides smoothly and decays to a halt.
4. **Intentional Cursor Freezing:** When a jaw clench or signal drop is detected, the pipeline intentionally zeroes out the Kalman velocity. This ensures the cursor remains perfectly locked on your target while you bite down to click.

📂 Architecture & Data Flow

```
Emotiv EPOC X -> [Bluetooth] -> Emotiv Cortex API
(wss://localhost:6868)
                       |
                       v
             [ cortex_client.py ]  (Async WebSocket ingestion)
                       |
           125ms Windows @ 256Hz  (Hop 62.5ms)
                       |
                       v
              [ pipeline.py ]
               /             \
    [ artifacts.py ]        [ baseline.py & csp.py ]
    - Check SNR             - Extract Mu/Beta power
    - Detect Jaw Clench     - Adaptive Z-Score Normalization
    - Flag Freezes          - Project 2D Intent
               \             /
                v           v
              [ kalman.py & mapper.py ]
              - Apply gain & deadzones
              - Smooth via Kinematic decay
```

```
         |
         v
[ run_live.py ]
- PyAutoGUI Move/Click
- Telemetry Logging
```

⚙️ Setup & Installation

1. Prerequisites

- **Emotiv Launcher / Cortex Service:** Must be running in the background.
- **Pro License:** Required by Emotiv to access the raw "eeg" data stream.
- **Python 3.10+**

2. Install Dependencies

```
pip install -r requirements.txt
```

3. Configure Credentials

Copy .env.example to .env and insert your API keys from the Emotiv Developer Portal:

```
CORTEX_CLIENT_ID=your_client_id_here
CORTEX_CLIENT_SECRET=your_client_secret_here
```

🚀 Usage Guide

This system operates in four distinct phases. **You must run these scripts as modules from the root directory.**

Step 1: Calibration (record_calib.py)

Records your baseline brainwaves to train the spatial filters.

```
python -m epoc_cursor.apps.record_calib
```

- **Instructions:** Follow the terminal prompts. Imagine moving the cursor **LEFT** for 30 seconds, then **RIGHT** for 30 seconds. Keep your jaw completely relaxed during this phase so muscle noise doesn't corrupt the EEG training data.

Step 2: Training (train_csp.py)

Computes the Common Spatial Pattern (CSP) weights based on your calibration data.

```
python -m epoc_cursor.apps.train_csp
```

- **Result:** Generates data/csp.json. This file maps your unique brain patterns to 2D screen coordinates.

Step 3: Live Control (run_live.py)

Starts the live BCI control loop.
`python -m epoc_cursor.apps.run_live`

- **Navigation:** Imagine the directions you trained.
- **Clicking:** Clench your jaw (bite down briefly). The cursor will freeze in place and execute a left-click.
- **Exit:** Press Ctrl+C in the terminal to safely close the session.

Step 4: Review & Tune (run_replay.py)

Analyzes the telemetry saved from your live session.
`python -m epoc_cursor.apps.run_replay`

- Use this to see how often the click triggered or if your cursor velocities are hitting the maximum clamps.

🎛️ Tuning Guide (config.py)

No two brains (or skulls) are exactly alike. You will likely need to adjust epoc_cursor/config.py for optimal performance.

Jaw Clench Sensitivity

- **Parameter:** CFG.CLENCH_VAR_THRESH (Default: 1800.0)
- **Too Sensitive?** (Clicking when you talk or swallow) -> Increase to 3000.0 or 4000.0.
- **Too Hard?** (Have to bite painfully hard to click) -> Decrease to 1000.0 or 800.0.

Cursor Speed & Jitter

- **Parameter:** CFG.GAIN (Default: 6.0) -> Increase for a faster cursor.
- **Parameter:** CFG.DEADZONE (Default: 0.35) -> Increase if the cursor drifts while you are resting.
- **Friction:** Inside kalman.py, the self.decay = 0.85 variable controls cursor "ice". Lowering it to 0.5 makes the cursor stop instantly when you stop thinking. Raising it to 0.95 makes it glide.

⚠️ Troubleshooting

1. **"Access not granted / Emotiv Launcher hangs"** The first time you run the script, the Emotiv Launcher application will pop up a window asking you to "Approve" the script's access. You must click Approve.
2. **"Stream established, but cursor isn't moving"** Check sensor contact. The artifacts.py module freezes the cursor if max_amp > 800 or std_dev < 0.1. Ensure your EPOC X felt

pads are thoroughly soaked in saline and that the reference sensors (black nodes behind the ears) have excellent skin contact.

3. **Cursor pulls heavily to one side** This means the baseline has drifted due to a loose sensor or drying saline. Restart run_live.py to reset the Adaptive Baseline normalizer, or slightly increase CFG.NORM_ALPHA so the system adapts to changes faster.